

Q17) Write a C program for DES algorithm for decryption, the 16 keys (K1, K2, ..., K16) are used in reverse order. Design a key-generation scheme with the appropriate shift schedule for the decryption process.

Aim: To write a C program to generate the **16 DES round keys** and display them in **reverse order** for the decryption process.

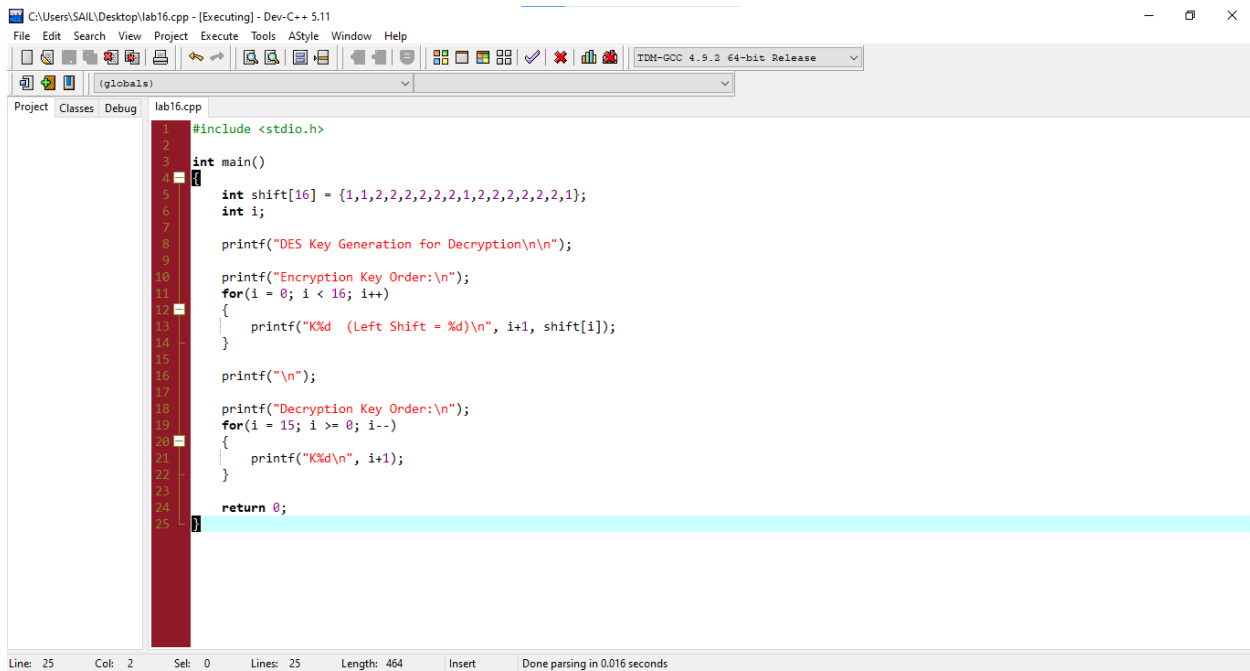
Algorithm:

1. Start the program.
2. Read the 16-character hexadecimal DES key from the user.
3. Convert the hexadecimal key into binary format.
4. Apply the **PC-1 (Permuted Choice-1)** table to obtain the 56-bit key.
5. Split the 56-bit key into two 28-bit halves (**C** and **D**).
6. Perform left circular shifts according to the DES shift schedule for all 16 rounds.
7. Apply the **PC-2 (Permuted Choice-2)** table to generate the 48-bit round keys **K1 to K16**.
8. Store all generated round keys.
9. Display the round keys in **reverse order (K16 to K1)** for DES decryption.
10. Stop the program.

Result:

Thus, the C program to generate the **DES decryption key schedule** was successfully executed. It was observed that the decryption process uses the same 16 round keys generated during encryption, but they are applied in the **reverse order (K16 to K1)**. This reverse key schedule enables DES decryption to recover the original plaintext correctly.

Code:



```
#include <stdio.h>

int main()
{
    int shift[16] = {1,1,2,2,2,2,2,2,1,2,2,2,2,2,1};
    int i;

    printf("DES Key Generation for Decryption\n\n");

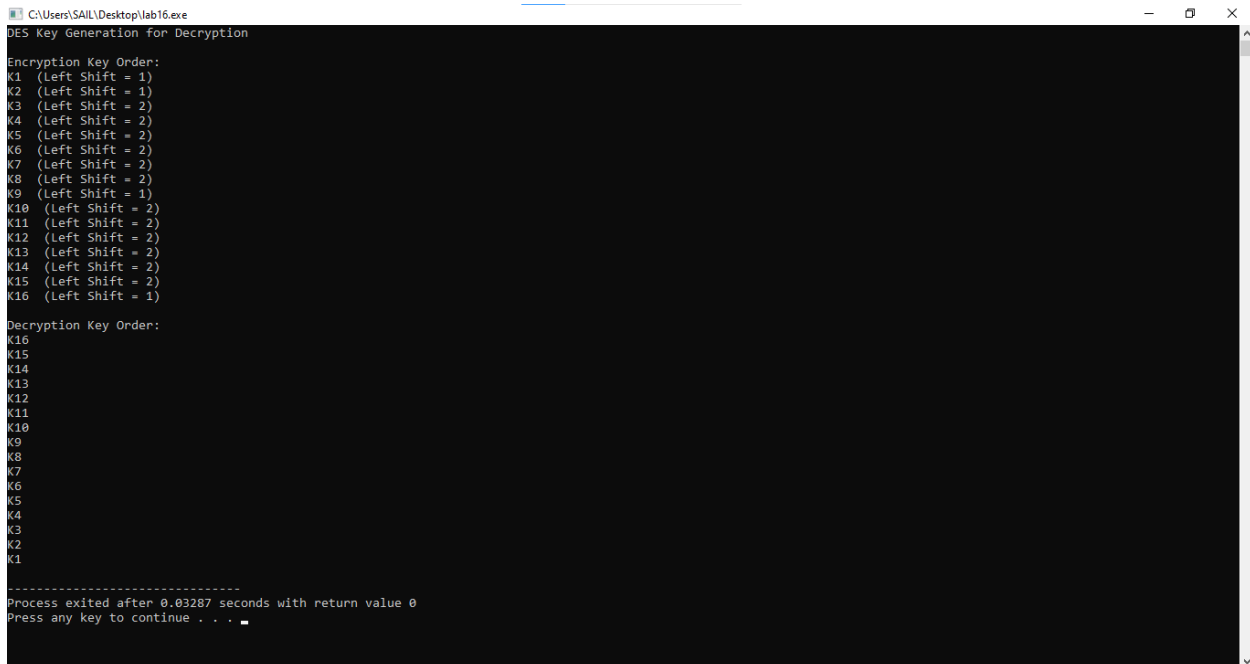
    printf("Encryption Key Order:\n");
    for(i = 0; i < 16; i++)
    {
        printf("K%d (Left Shift = %d)\n", i+1, shift[i]);
    }

    printf("\n");

    printf("Decryption Key Order:\n");
    for(i = 15; i >= 0; i--)
    {
        printf("K%d\n", i+1);
    }

    return 0;
}
```

Output:



```
DES Key Generation for Decryption

Encryption Key Order:
K1 (Left Shift = 1)
K2 (Left Shift = 1)
K3 (Left Shift = 2)
K4 (Left Shift = 2)
K5 (Left Shift = 2)
K6 (Left Shift = 2)
K7 (Left Shift = 2)
K8 (Left Shift = 2)
K9 (Left Shift = 1)
K10 (Left Shift = 2)
K11 (Left Shift = 2)
K12 (Left Shift = 2)
K13 (Left Shift = 2)
K14 (Left Shift = 2)
K15 (Left Shift = 2)
K16 (Left Shift = 1)

Decryption Key Order:
K16
K15
K14
K13
K12
K11
K10
K9
K8
K7
K6
K5
K4
K3
K2
K1

-----
Process exited after 0.03287 seconds with return value 0
Press any key to continue . . .
```